2027 EDISON AWARDS

NOMINEE

PUBLIC SAFETY · SECURITY · DIGITAL INTEGRITY

FOR MILLENNIA

CIVILIZATION & ECONOMIC LIFE

Trust — the oldest question of civilization.

How can we verify before we act?



SMVRS™

INDEPENDENT VERIFY

Proof Before Effect.

Portable semantic proof across devices and locations.

SIGNED RECORD
- Identity & Authority
- Evidence
- Events & Time
- Policy & Rules

TWO INDEPENDENT VERIFIERS
- Python Reference
- Node.js Reference
- Same Input Same Rules

EXPLAINABLE RESULT
- Verified / Failed
- Explicit Reason Codes
- Detects Tampering
- Consistent Results

FOR EVERYONE · EVERY FAMILY · EVERY ENTERPRISE · EVERY NATION



INDIVIDUALS

Personal records, identity and rights



FAMILIES

Health, education, assets and services



ENTERPRISES

Contracts, compliance and trusted operations



INSTITUTIONS

Public services, regulation and trusted digital state



NATIONS & REGIONS

Interoperable trust across jurisdictions

Inventor: Dr. FURONG BEI · SMVRS.com · ATMS.com · contact@atms.com

THE QUESTION

Trust — civilization's millennia-old conundrum.

For thousands of years, economic life has depended on records, and people have largely had to rely on whoever held them. Trust is a foundation of civilization's economy. The question has always been the same: how can a receiver verify before acting?

c. 3200 BCE

Clay tablets record grain, debts and ownership.

1494

First printed exposition of double-entry bookkeeping; not its invention.

TODAY

Databases and digital signatures move records across systems, borders and machines.

SMVRS answer:

Proof Before Effect.

For compatible records, recipients with trusted keys and a pinned profile can recheck defined evidence. Using this before consequential action is the design goal, not an accepted production effect gate.

AT A GLANCE

SMVRS™ Independent Verify: portable proof with independent semantic verification.

For supported signed records, two separately built verifier paths can recompute selected evidence and policy-profile checks. Controlled Golden and evidence-tamper demonstrations show VERIFIED and FAILED results. Historical engineering records report cross-region re-verification; independent service auditing remains outstanding.

01 · CONCEPT

A valid signature supports integrity of signed content. SMVRS adds receiver-side checks of stated evidence, events, time and the pinned verification profile.

02 · VALUE

Python and Node.js verifiers, built separately, recompute the proof on the receiver's side. In controlled tests an intact record passed all 18 checks; altered evidence failed with the exact reason code. Standard cryptography, no special hardware.

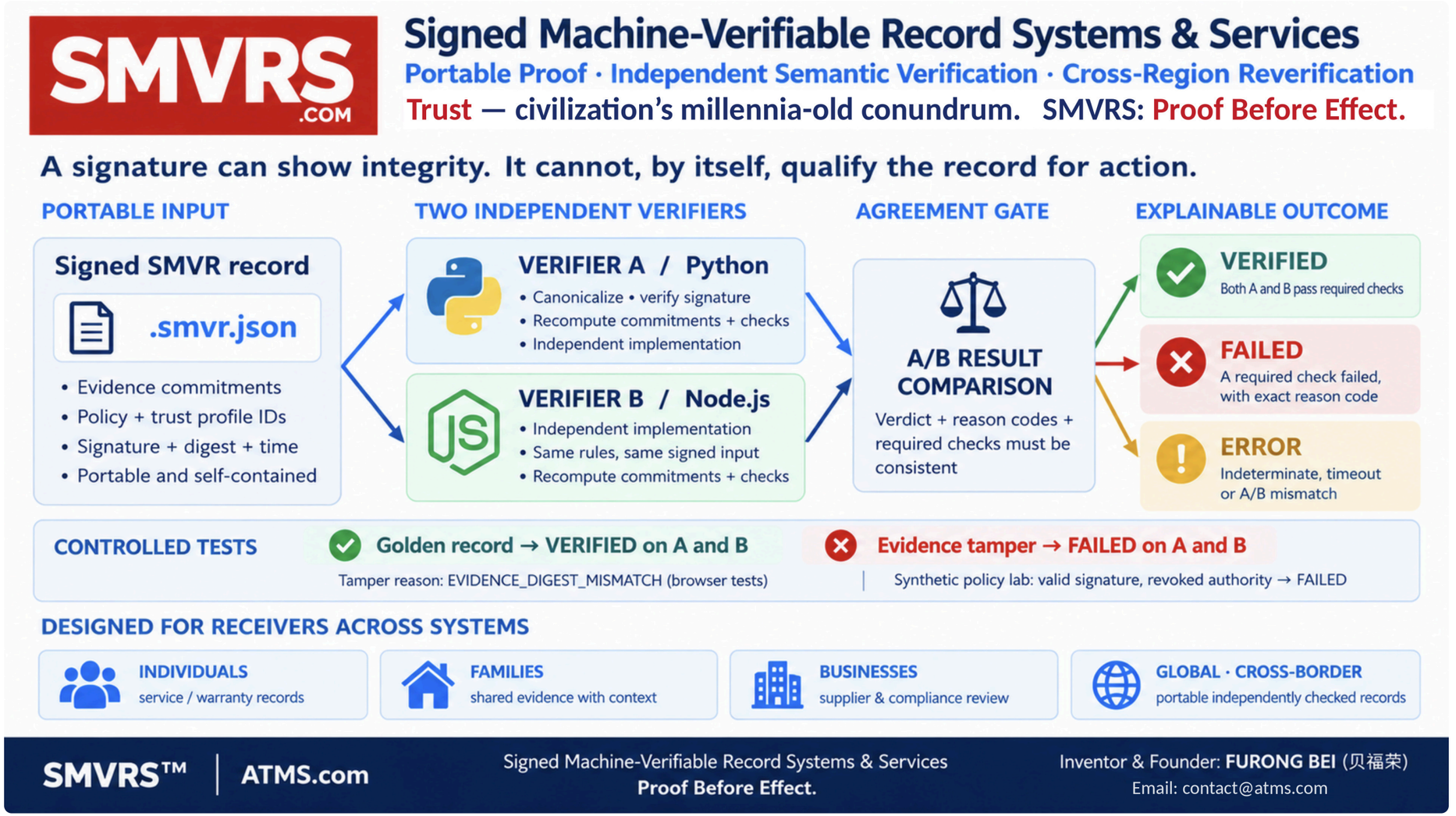
03 · IMPACT

Individuals, families, enterprises and institutions could use compatible portable proof. Broader public-safety and cross-border impact needs independent pilots and sector-specific acceptance.

> Check the evidence. Know the limits.

THE INNOVATION

How SMVRS Independent Verify works



Architecture: supported signed record to two verifier paths to a proposed agreement gate and typed result. The public demo does not establish a production action gate.

WATCH · READ · CHECK

Key claims, evidence and research boundaries

60-second film	youtu.be/HiTej_EingU
Edison nominee page	edisonawards.com/nominees/smvrs-independent-verify
Product	smvrs.com
Company	atms.com
Public beta	bei.app
Patent records (Justia)	patents.justia.com/inventor/furong-bei
Patent records (Google)	patents.google.com/?inventor=Furong+Bei
Contact	contact@atms.com

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The innovation · p. 4

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02 Value · pp. 16–20

03 Impact · pp. 21–23

Architecture · pp. 24–28

Patent applications · p. 29

Inventor and contact · p. 30

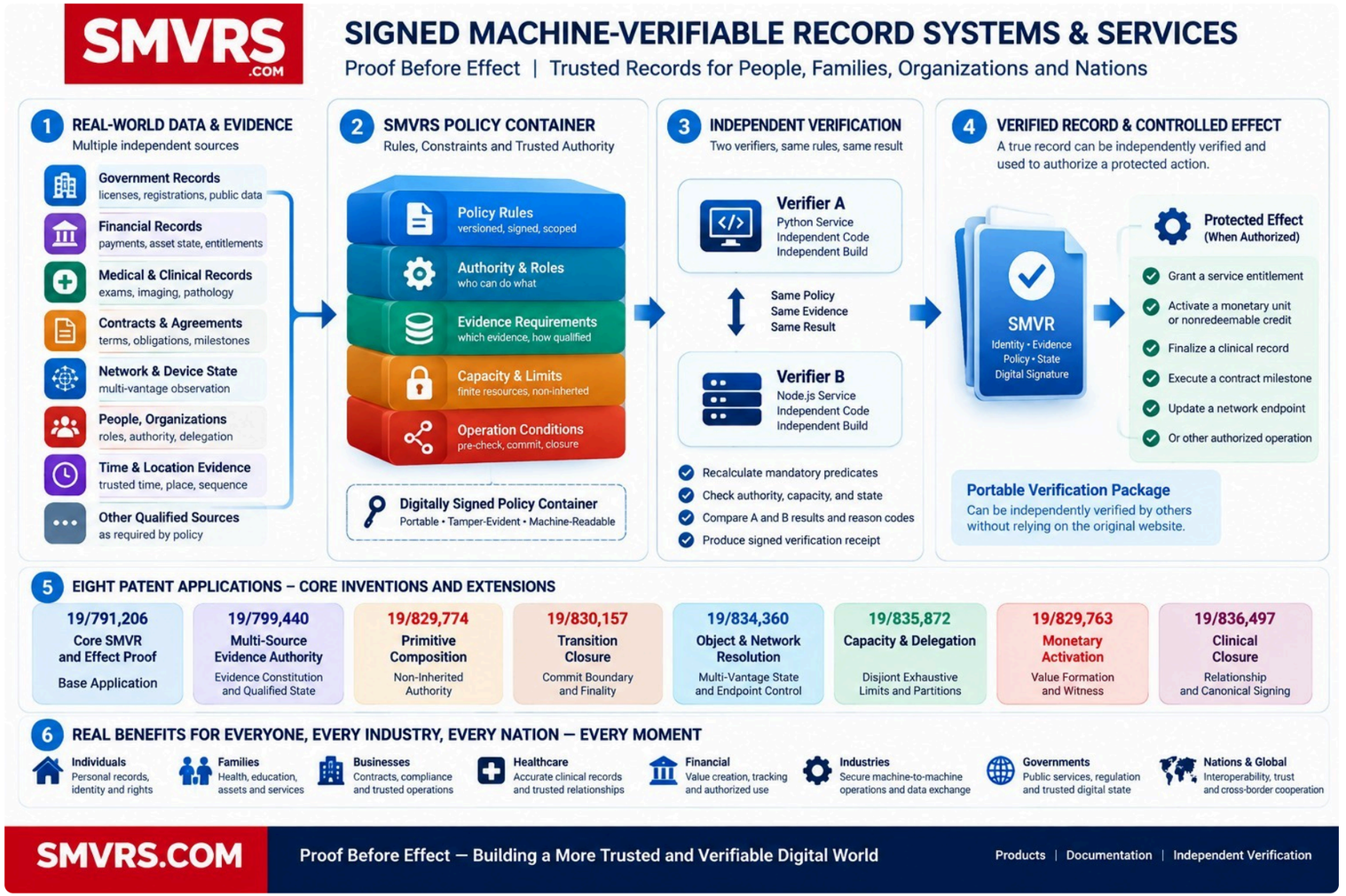
EVIDENCE STATUS & PUBLIC READER GUIDE

DEMONSTRATED / CONTROLLED	Two verifier browser paths: Golden record VERIFIED; evidence tamper FAILED with typed reason.
HISTORICAL / INVENTOR REPORTED	Cross-region engineering; 13 negative-vector classes; 79/79 synthetic local tests. Independent audit pending.
RESEARCH / NOT LIVE	Signed Policy Container, protected effect, financial/clinical results, global single-winner signing.

Current beta sites provide planning/catalog functions; do not infer live protected signing or action control.

Protocol BEI-SMVR-0.1 · Verification profile SMVR-VERIFY-1 · Golden record smvr:6d5845d693596ca36e0c0de5

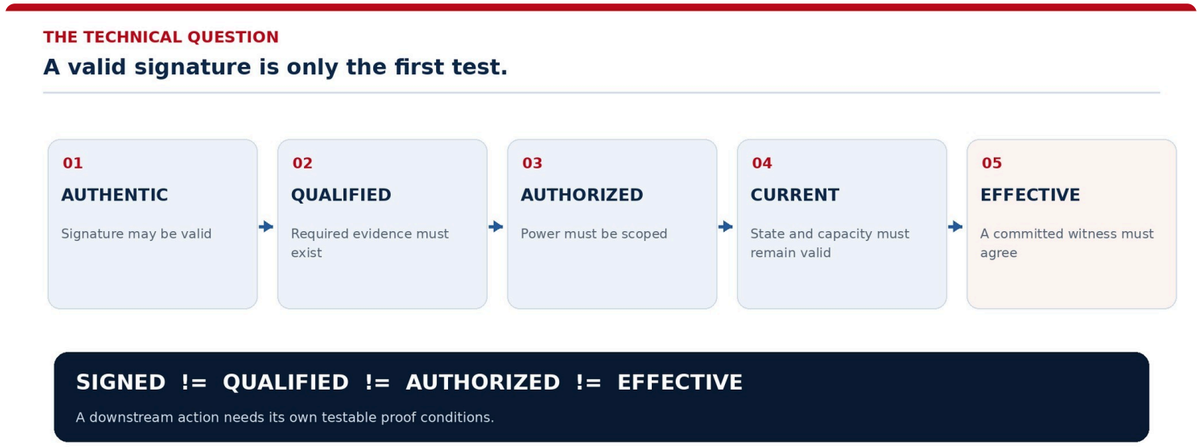
Core invention: from record to qualified action



Qualified real-world inputs, a signed policy, two independent verification paths, a portable record and a protected-effect boundary. The effect stages illustrate the broader architecture described in the patent applications.

A valid signature is the start of trust, not the whole of it.

Authentic bytes can still carry incomplete evidence, outdated authorization, a wrong relationship or a result that never took effect. SMVRS addresses this missing link in cross-system digital trust.



Machine-readable conditions

SMVRS separates cryptographic authenticity, semantic relationship validity, evidence sufficiency, bounded authority and the operation’s actual result.

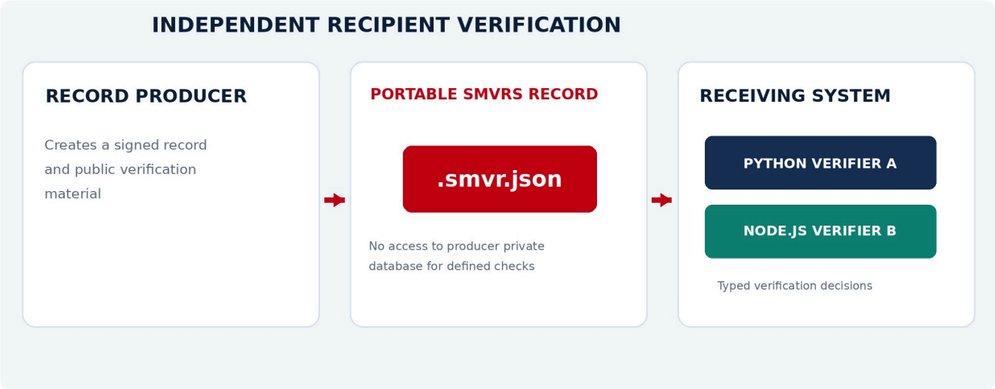
A falsifiable rule

A signed but incomplete or wrongly linked record fails the required check instead of inheriting trust from a logo, a dashboard, a copied source or a majority vote.

From trusting a producer’s claim to independently checking the conditions for relying on it.

A portable record can be checked away from the issuing system.

The SMVR record binds actor, behavior, evidence, policy, time and claimed decision. Independent Python and Node.js implementations recompute the validation rules and report typed failures.



Demonstrated in controlled tests

Golden-record acceptance; rejection of behavior and evidence tampering; semantic-binding cases; agreement between two independent verifiers.

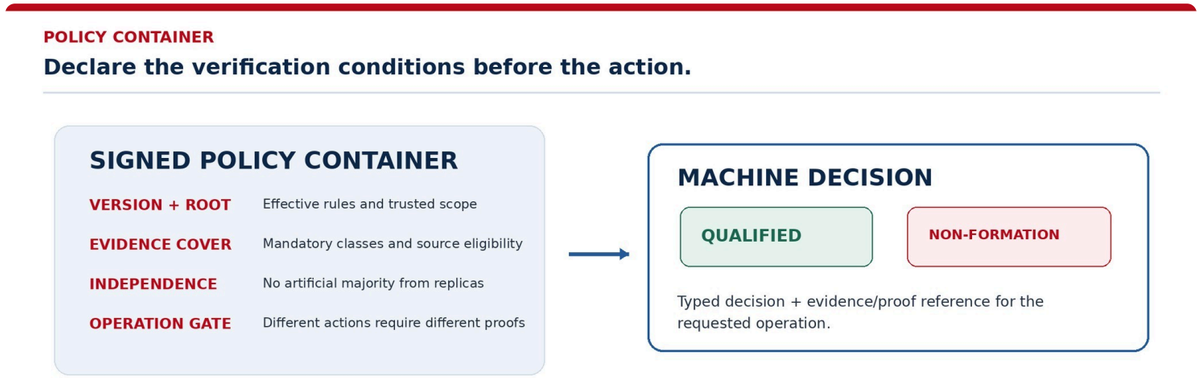
Beyond the signature

A valid signature supports integrity. Supported SMVR checks test declared relationships, but cannot alone establish external truth, live authorization or completed effect.

> Controlled record verification is demonstrated; production protection of real-world effects remains a separate development gate.

The applicable rule is pinned before the decision.

In the proposed architecture, a signed Policy Container would bind context, effective version, trust roots, required evidence, scope, operation and revocation. The current evidence-policy experiment uses synthetic test fixtures.



Why it matters

Without versioned rules, two services may verify the same bytes yet apply different authority, purpose, time or privacy requirements.

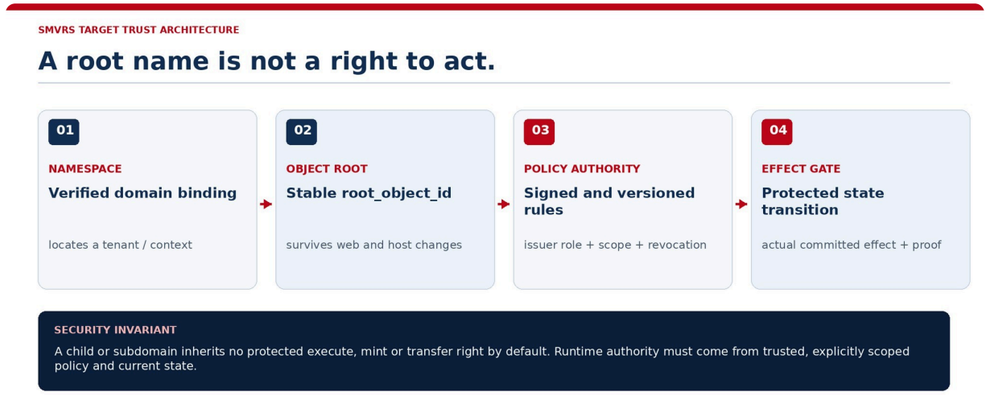
Prototype and roadmap

A signed evidence-policy experiment runs on synthetic data. A cross-service Policy Container registry and a non-bypass action gate are the next extensions.

> Policy qualification remains research, not a released production action gate.

A web address can locate a record. Only trusted policy can authorize an action.

The proposed authority model distinguishes a verified website from persistent object identity, issuer trust and scoped permissions. A child object would receive no protected execution right without an authenticated, current policy grant.



Stable identity beyond a URL

The model is designed to retain stable object identity when host or domain names change.

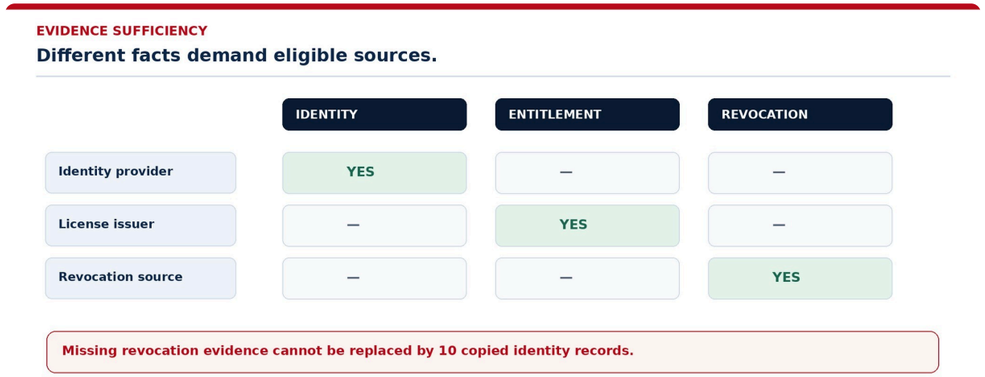
Fail-closed delegated rights

A parent signature or an inherited name cannot silently confer mint, transfer or execution privileges.

Names aid discovery; machine authority comes from authenticated trust, scoped grants and current-state checks.

Each required kind of evidence must come from an eligible source.

An operation can require evidence of identity, entitlement, revocation and observed effect, each from its own eligible authority. Correlated copies do not count as independent corroboration.



A sharper question

Instead of asking which database is most trusted, SMVRS asks whether the evidence classes this operation needs are complete and independently eligible.

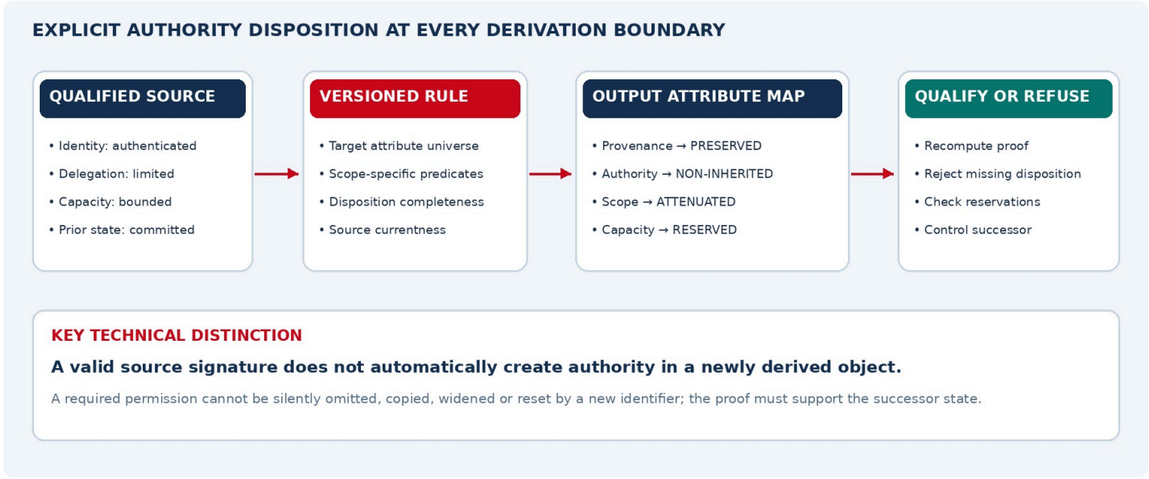
Tested on synthetic fixtures

Signed test fixtures accept a complete qualifying set and reject a missing mandatory class, an out-of-scope assertion or disallowed correlated origins.

> Designed to verify without one universal master database.

Provenance can persist. Permission is qualified again.

Primitive composition separates where a derived object came from and what protected authority it may exercise. Attribute disposition, allowed scope and live predecessor conditions constrain each successor state.



Scope-limited derivation

The proposed derivative-control rules would block unauthorized scope, quantity, duration or purpose expansion in a child or transformed object.

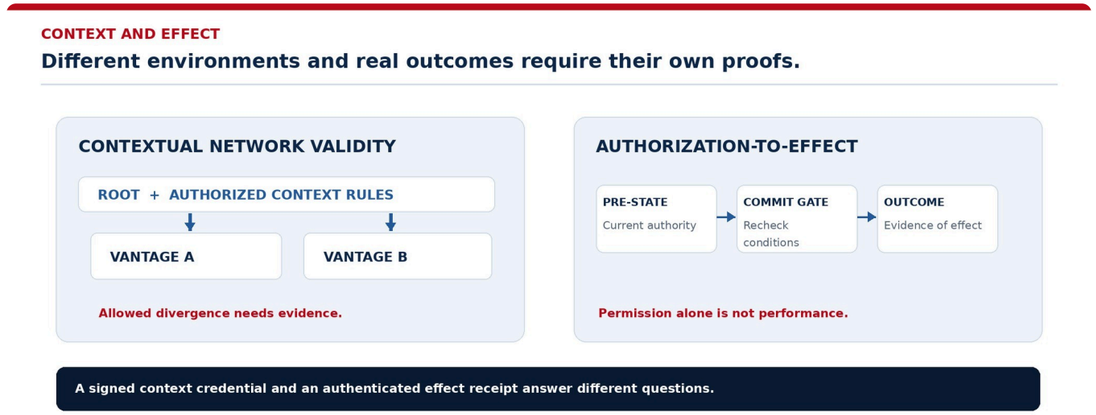
At the commit boundary

Transition closure rechecks current state, policy, replay and authorized effect when an operation is committed, not only when it is proposed.

> **Derived objects keep their history but must earn their authority.**

Correct observations can differ. A claimed effect must be shown.

Rooted multi-vantage resolution qualifies differences between observations under an authenticated policy. The foundational application addresses the match between authorized preconditions and demonstrated state effects.



Contextual verification

A network difference can be permissible under policy; a failing mandatory observation cannot be outvoted by a generic majority.

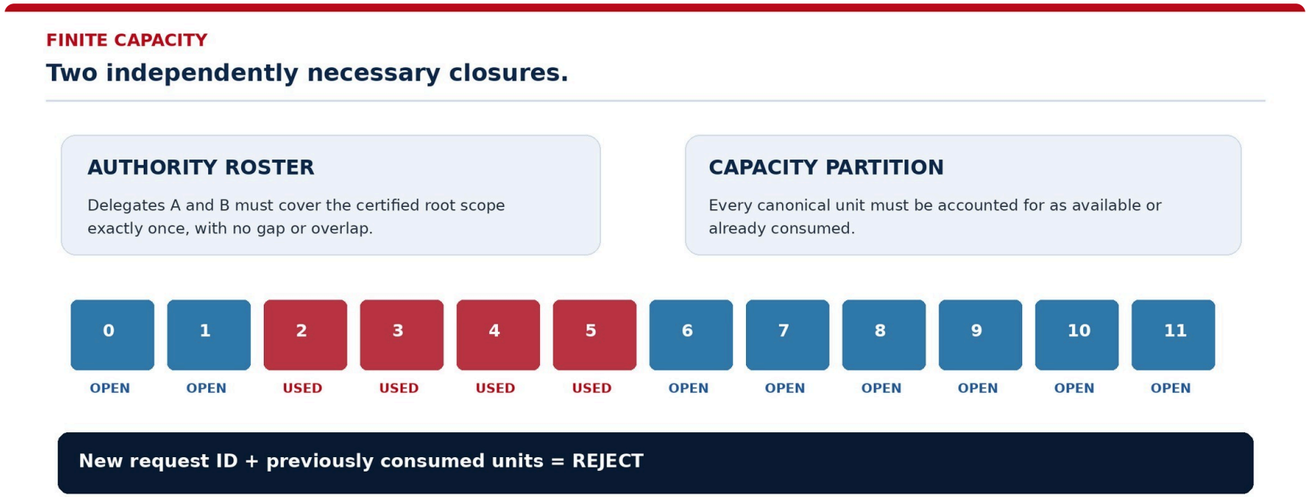
Effect correspondence

A proposed effect receipt would bind actor, policy, predecessor state, effect evidence and successor state. Production commit enforcement is not yet accepted.

> Executed-effect proof remains a separate research and release gate.

An authorized delegate must not spend the same unit twice.

Certified capacity requires two complete checks: who is entitled to allocate units, and which exact units remain available. Both must hold for a serialized, witnessed successor state.



Laboratory capability

Test-key signatures, a complete authorized roster, finite-unit accounting, a local serialization point and detachable historical witness checks.

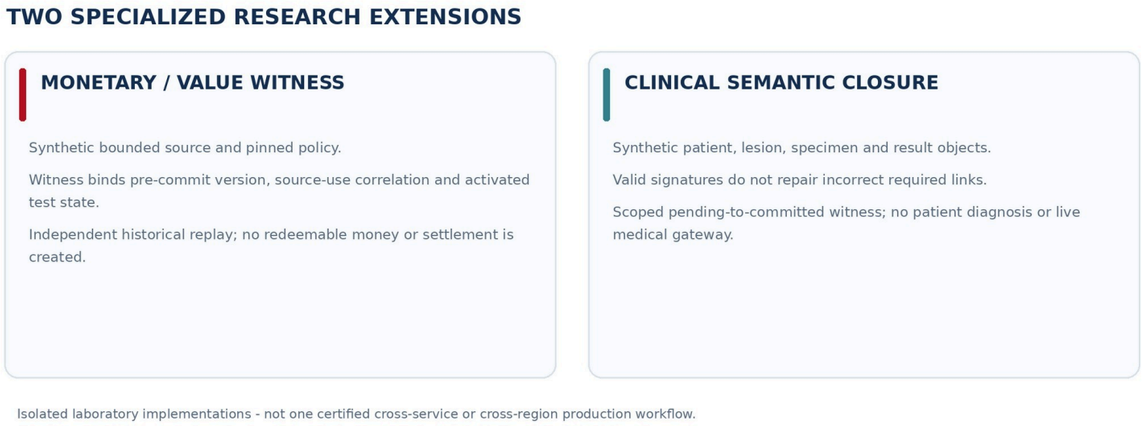
Anti-regeneration

Research goal: prevent changed aliases, splits or request identifiers from making consumed units available again. Global concurrency proof is still pending.

> Local capacity tests do not prove global single-winner issuance.

The same rules, tested in two demanding domains.

Research adapters apply SMVRS principles to bounded test-value activation and to typed clinical-record relationships, each as a separate technical extension.



Bounded test-value activation

A simulated monetary controller checks pinned policy and version, source-use correlation, commit-time witness and historical validity, using simulated values only.

Clinical relationship closure

Synthetic patient, specimen and result graphs expose incorrect signed links and distinguish a candidate from a committed record, supporting data integrity rather than diagnosis.

Narrow, repeatable machine checks that each sector can adopt on its own terms.

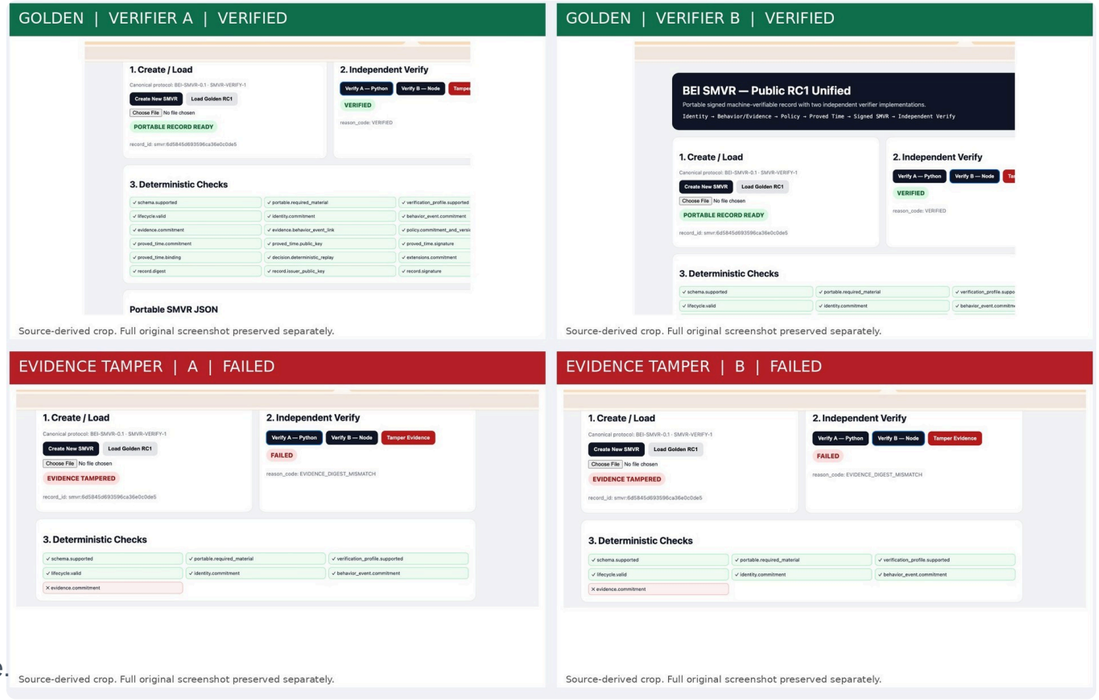
Same record, two verifiers, the same answer.

INPUT	VERIFIER A · PYTHON	VERIFIER B · NODE.JS
Synthetic signed Golden record	VERIFIED · 18 checks	VERIFIED · 18 checks
Synthetic evidence tamper	FAILED	FAILED

Reason code on both verifiers: `EVIDENCE_DIGEST_MISMATCH` at `evidence.commitment`.

Record: `smvr:6d5845d693596ca36e0c0de5`

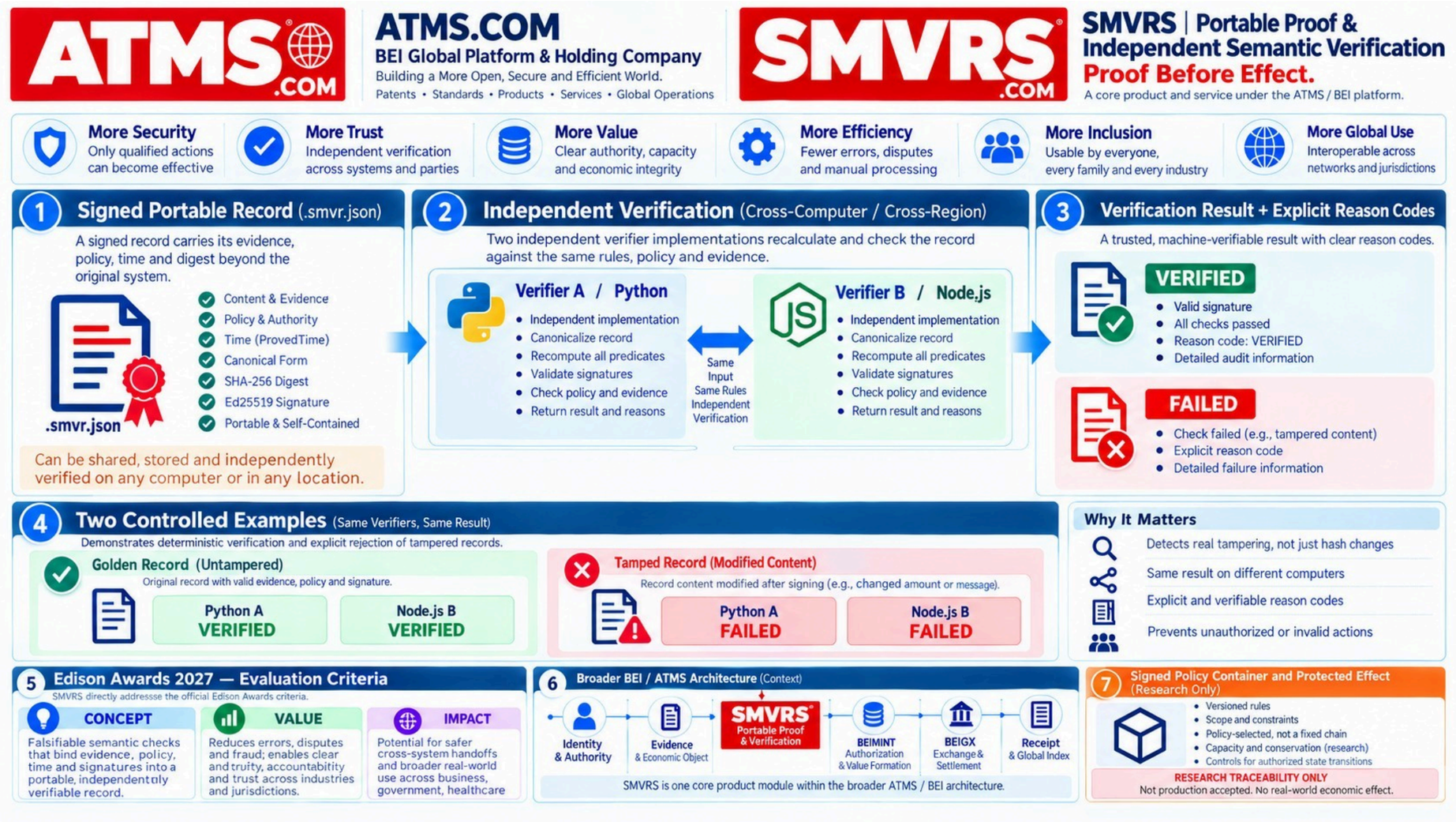
Historical engineering records report Oregon-to-Northern-California record verification and tamper rejection. Public browser screenshots do not establish the region of each request.



Scope: historical inventor-run test results, not a third-party security audit or universal correctness guarantee.

Controlled browser tests on synthetic signed records, October 2026.

Four controlled cases mapped to the Edison criteria



Golden record VERIFIED on A and B; evidence tamper FAILED on A and B, each with its exact reason code.

A clear first customer test: software-license allocation.

An Economic Node with ten non-redeemable service seats offers a falsifiable first use case. A requester may hold a valid credential yet lack entitlement, revocation coverage or unused licensed capacity.



Useful failure cases

Proposed customer test: reject missing entitlement, stale rights, ineligible corroboration, unauthorized delegation and repeated use of the same seats.

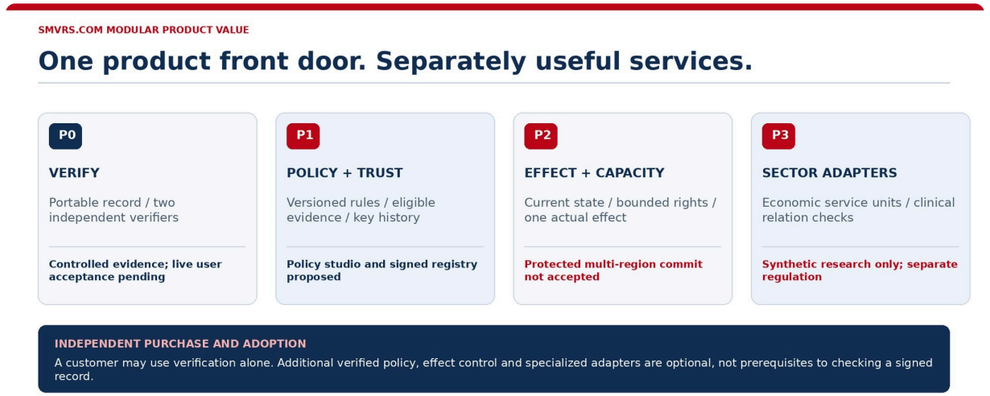
Practical package

Portable verification is a distinct service candidate. Evidence and finite-capacity controls remain separate laboratory modules; no complete live customer flow is claimed.

Value is measured by the decisions a system can independently justify.

Independent verification first. Deeper assurance when an operation needs it.

SMVRS.com presents a phased service architecture: portable Verify first; signed Policy, eligible Evidence, protected Effect and industry adapters only after separate acceptance.



Near-term customer outcome

A recipient uploads a signed record, sees two independently computed results and exports the comparison.

Longer-term customer outcome

Organizations add signed policy qualification or scoped effect control as their security, operational and sector requirements call for it.

Each edition is useful on its own; none requires adopting the monetary or clinical modules.

Evidence is stronger when failure is testable.

2

independent verifier implementations

Python and Node.js, built separately from one specification.

13

archived negative vector classes

Each kind of failure is preserved as a repeatable test.

79/79

Inventor-reported synthetic lab tests

Policy, evidence and capacity fixtures, run on synthetic data.

Cost

Standard cryptography (SHA-256, Ed25519), no special hardware, designed for integration through APIs with software, networks, machines and IoT devices.

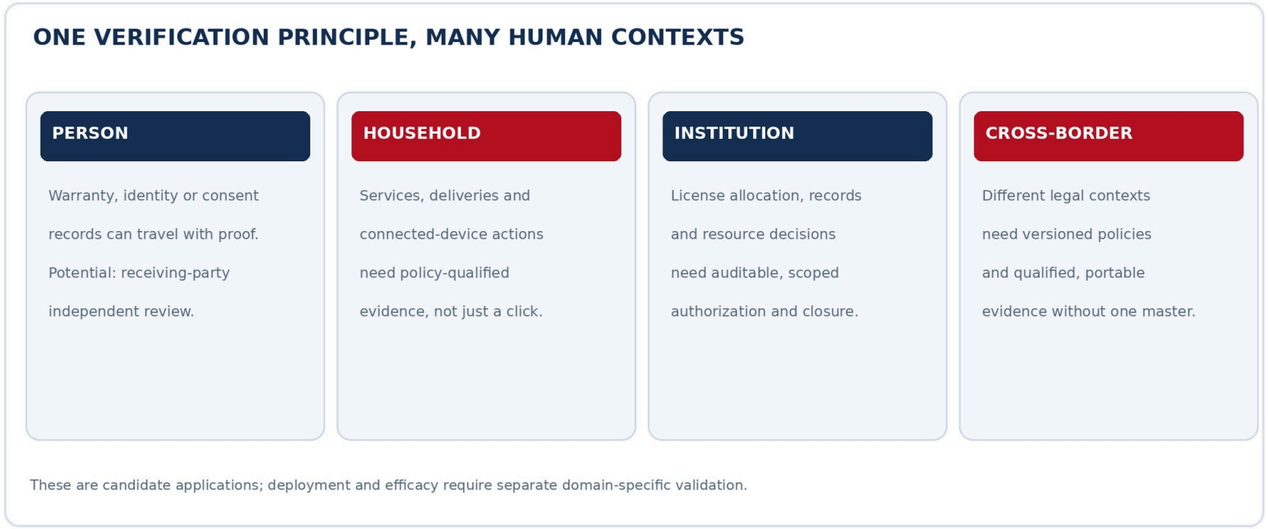
Benefit

Controlled tests reject a changed evidence commitment with a named reason. Preventing release of money, care or goods requires a separately verified production action gate.

> Cost and real-world benefit require independent measurement.

Built for people, families, institutions and nations.

A recipient checks a digital assertion against its applicable rules, rather than depending on the producer’s success label alone.



Everyday scale

A person, household or small business receives a portable result and verifies who asserted it, which evidence and policy apply, and whether the record is current.

Available planning beta: 365 industry labels / 999 need templates (not live industry integrations).

Across systems and borders

Institutions and regional services exchange proofs without assuming a single database or universal authority, each applying its own law, privacy and standards.

> Portable proof needs compatible systems, trusted keys and local safeguards.

A layered service each industry can adopt at its own pace.

The SMVRS roadmap starts with portable verification and may add evidence qualification, signed policy, controlled effects and sector modules after separate engineering and regulatory reviews.



A shared architectural direction; today, only the verifier and separate local research components have controlled tests.

Immediate service

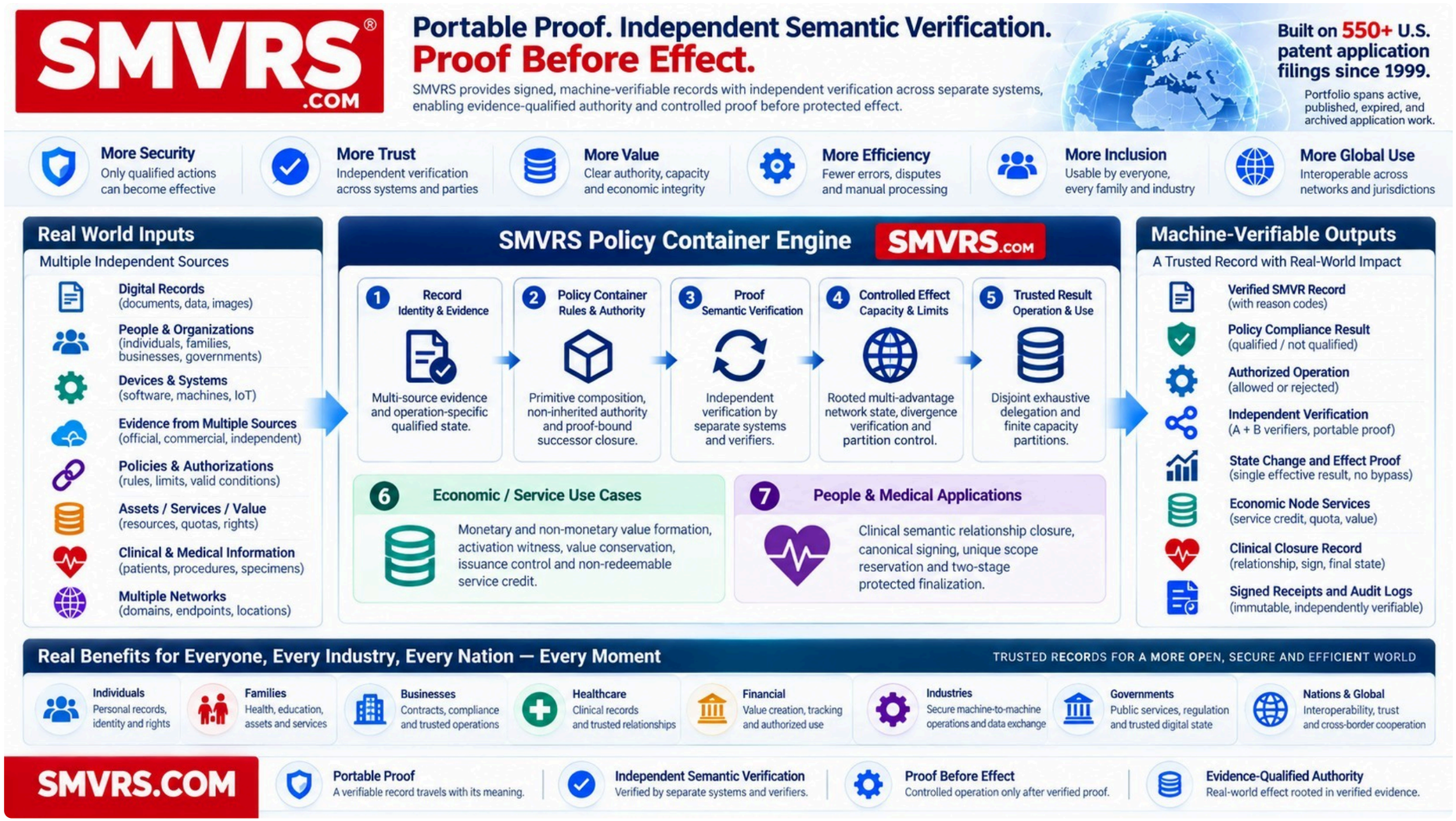
Record receipt, detached check, explicit result and reason code, and evidence export for institutional handoffs.

Longer-term services

Scoped policy decisions, capacity certificates, signed effect receipts and adapters for software, enterprise records, devices and regulated systems.

> A possible future safety layer for AI automation, subject to deployment and effectiveness testing.

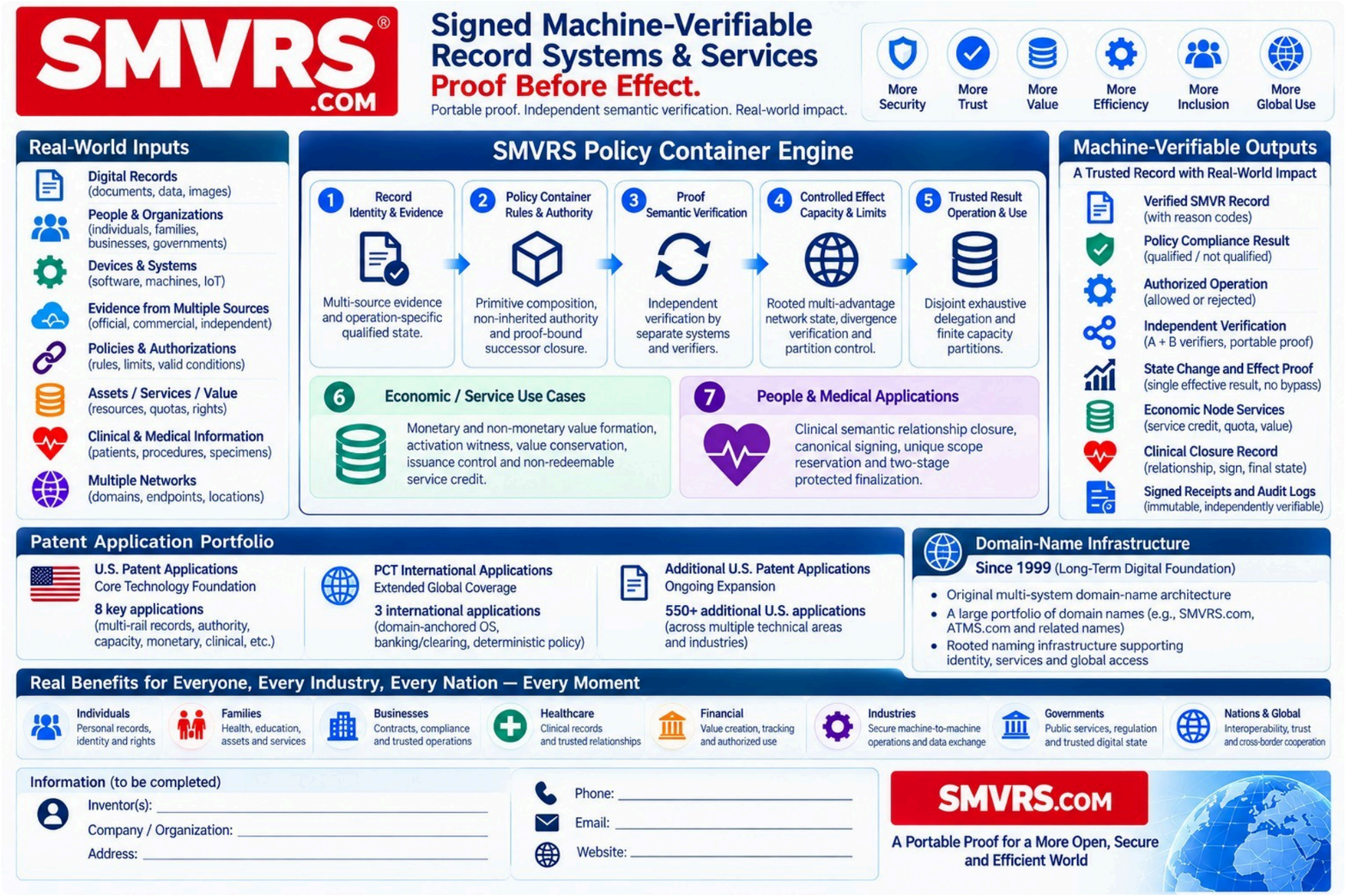
Proof Before Effect: reach and long-term roots



Archival inventor diagram: "1999" refers to domain/digital infrastructure work, not patent filing dates. Portfolio counts are not issued-patent counts.

ARCHITECTURE · FIGURE 2

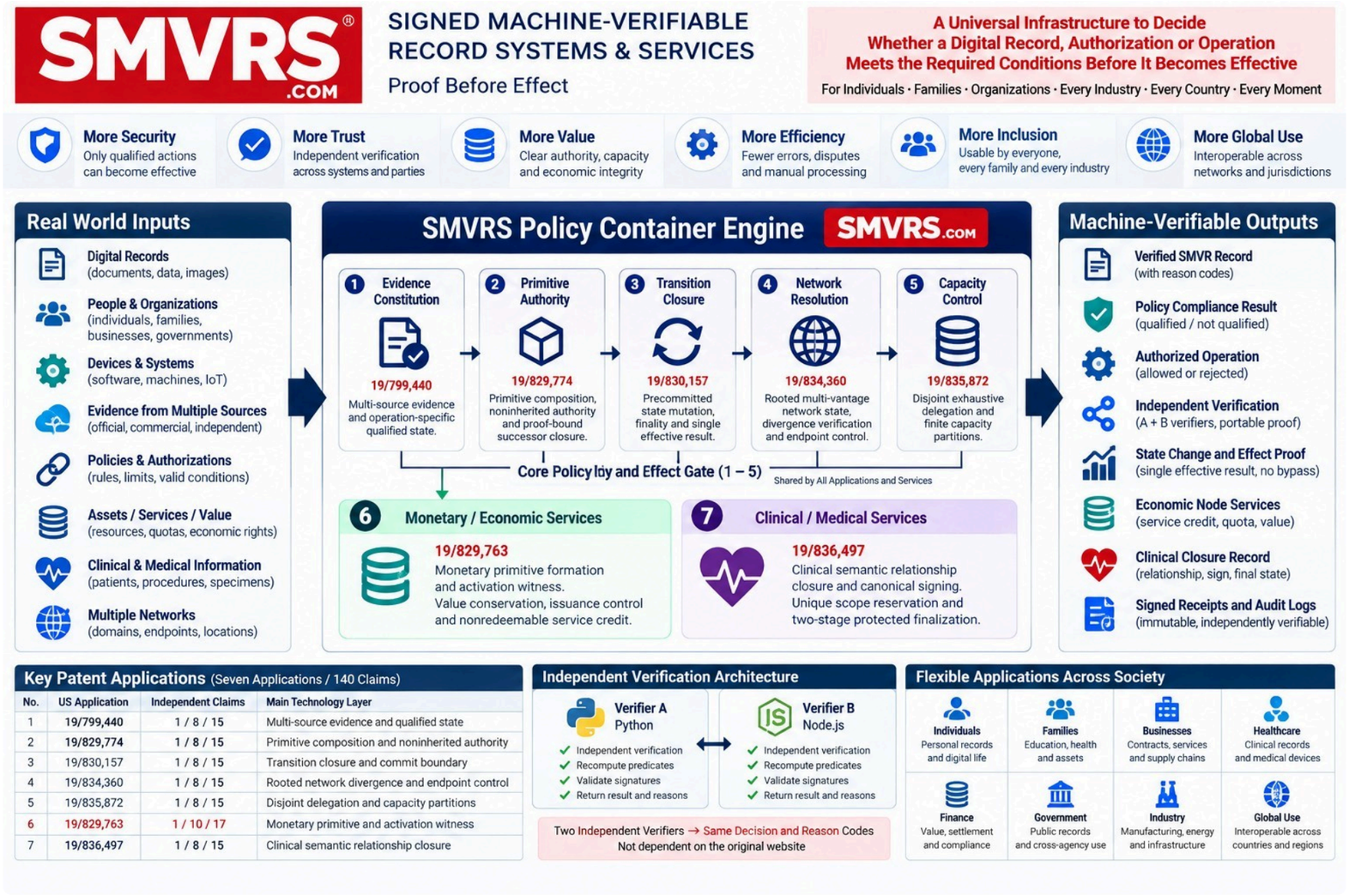
Complete product, patent and domain landscape



Policy Container Engine, economic and clinical extensions, patent application portfolio, rooted-domain assets and application areas.

ARCHITECTURE · FIGURE 3

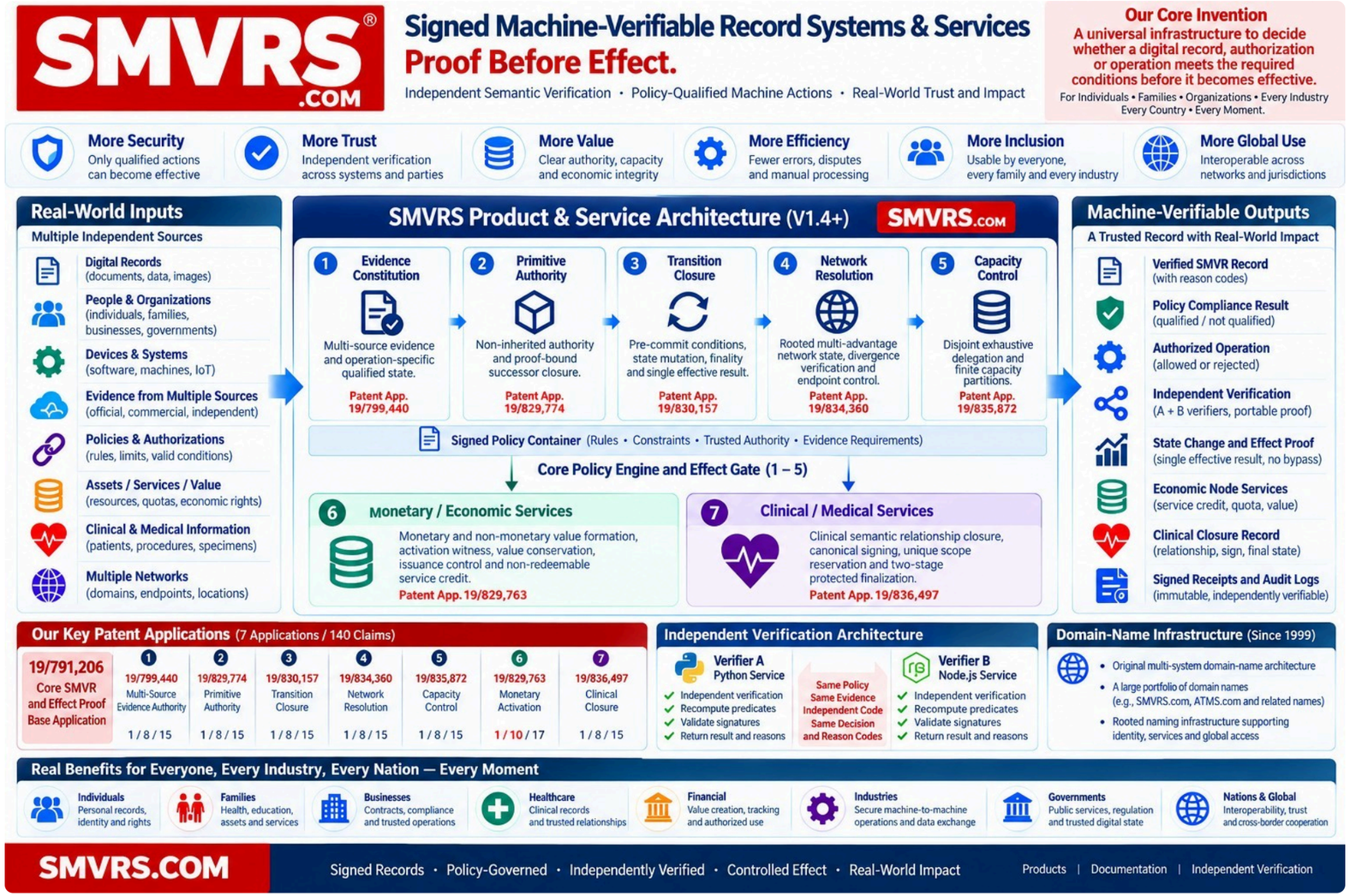
Seven technical layers and independent verification



Seven technical application layers with separate A/B verifier roles and industry inputs and outputs. Application numbers identify filings.

ARCHITECTURE · FIGURE 5

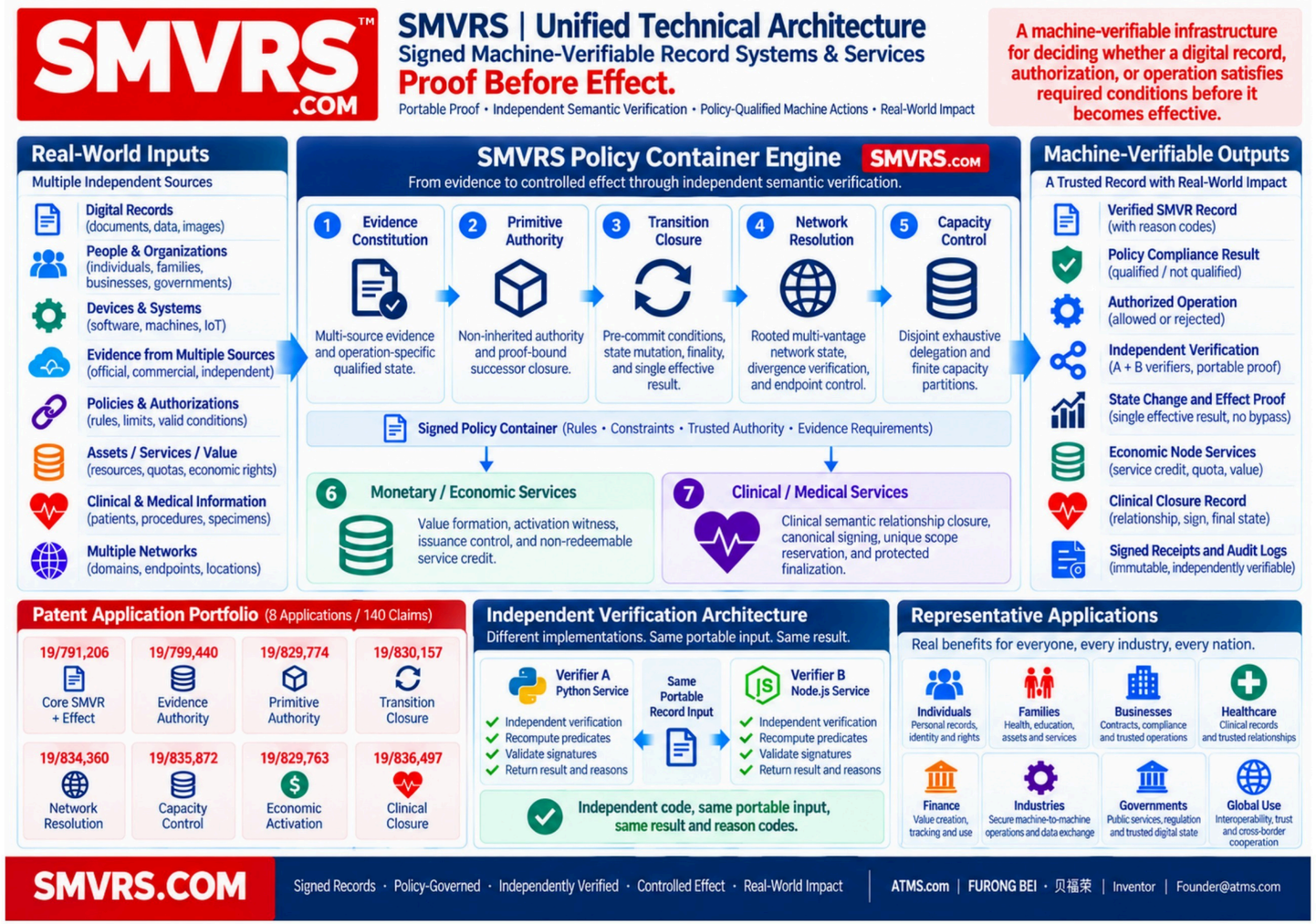
Detailed patent-core and industry blueprint



Evidence, primitive composition, transition closure, rooted network resolution, capacity, and economic and clinical specializations.

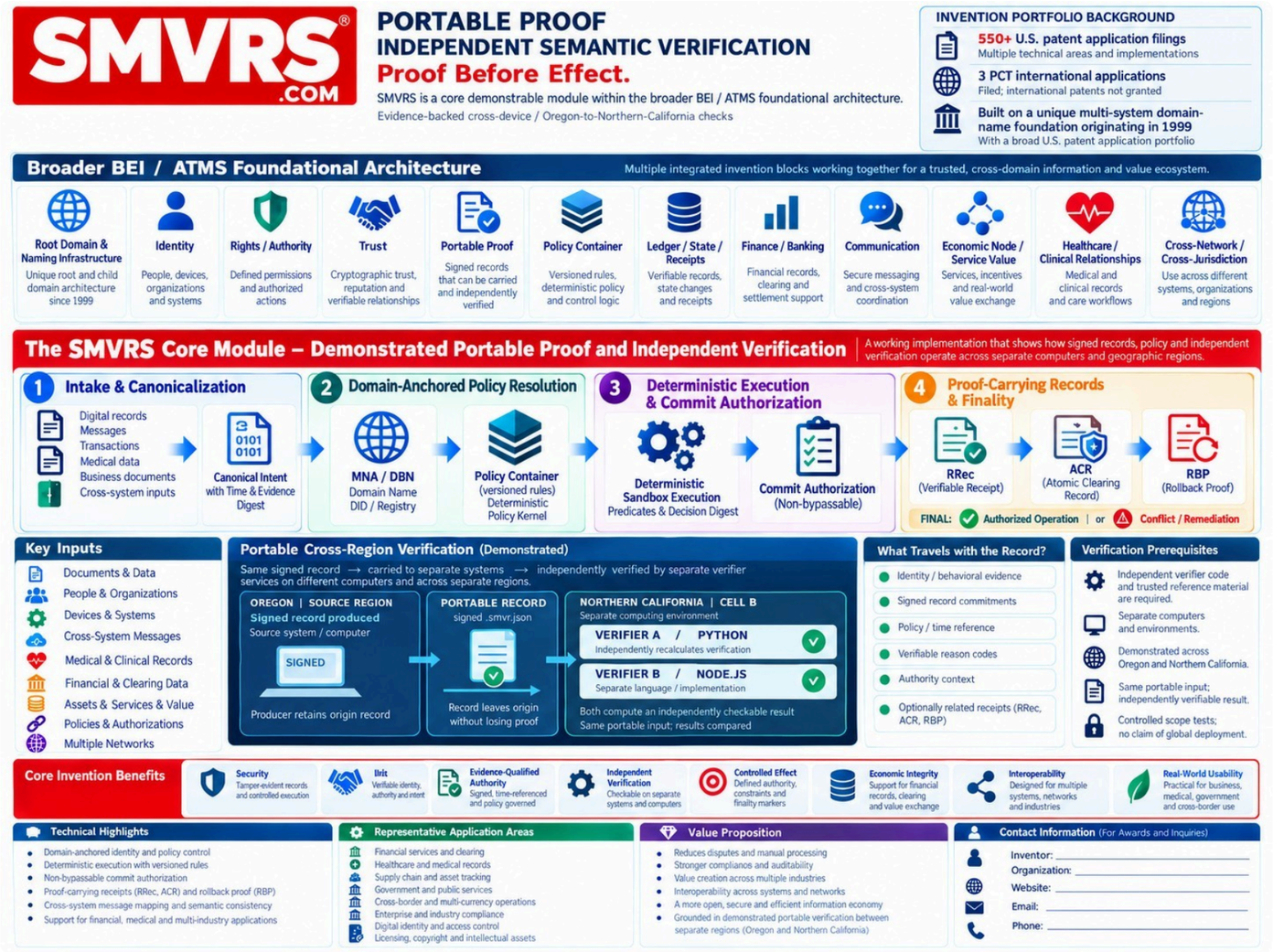
ARCHITECTURE · FIGURE 6

Eight related applications, one controlled system



Signed policy, five technical layers, economic and clinical extensions, and dual verifiers.

BEI × ATMS foundation, portable receipts and future extensions



Inventor research illustration of rooted-domain identities and portable receipt concepts within the broader BEI × ATMS architecture.

PATENT APPLICATIONS

Eight related U.S. applications, distinct technical roles.

19/791,206	Authorization–Effect Correspondence
19/799,440	Evidence-Scoped State Formation
19/829,774	Primitive Composition / Non-Inherited Authority
19/830,157	Transition Closure / Commit Boundary
19/834,360	Multi-Vantage Contextual Resolution
19/835,872	Certified Capacity / Dual Closure
19/829,763	Monetary Control / Activation Proof
19/836,497	Clinical Semantic Relationship Closure

556

Application records in inventor-supplied USPTO list (Oct. 12, 2026); mixed types and legal statuses.

Across the broader BEI x ATMS portfolio. Eight cited U.S. filing references concern related research topics; current claims and legal status require separate verification.

Since 1999

A domain-rooted naming and digital-infrastructure initiative, developed by the inventor and self-funded.

Portfolio note (public disclosure): 556 denotes inventor-supplied USPTO application records, not 556 issued patents or 556 SMVRS patents. The 1999 date is NOT a patent-filing date. The eight listed references are not proof of grant, patentability or production deployment.

Public records: patents.justia.com/inventor/furong-bei · patents.google.com/?inventor=Furong+Bei

2027 EDISON AWARDS

NOMINEE

SIGNED MACHINE-VERIFIABLE RECORD SYSTEMS & SERVICES



SMVRS™

INDEPENDENT VERIFY

Proof Before Effect.

TRUST — CIVILIZATION'S MILLENNIA-OLD CONUNDRUM

How can we independently verify before we act?

FOR EVERY PERSON | EVERY FAMILY | EVERY NATION | EVERY REGION

Toward a verifiable foundation for civilization and economic life.

Inventor: Dr. FURONG BEI

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